

DL for ENG

## LECTURE 11.2

# Dynamics and Energy

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MODEL · GRIP · JOULES



*The same car, now as a dynamical system. A bicycle model with one parameter describes its motion, the tyres set a hard ceiling on its speed through every corner, and a lap costs joules in winding heat, rolling and the computer's draw. The lecture identifies the model from a coast-down and finds the cheapest lap with a network whose last layer cannot break the grip limit.*

# Learning Objectives

## SECTIONS, IN ORDER

- 1 **The vehicle** — a bicycle model, and the grip limit it must respect
- 2 **The energy** — where a lap's joules go, and the cheapest lap
- 3 **Identify and optimise** — a coast-down, and the limit as a layer

*Liu carries the inverse fit and the optimisation through a network. Every car number in the study notebook is a placeholder of the right order, marked as such; the on-car notebook measures your own, and the car has no encoder, so its speed comes from lap timing or from the camera.*

## READ, AND WHAT EACH ANSWERS

|           |                                 |
|-----------|---------------------------------|
| Liu ch. 4 | what can a coast-down identify? |
| Liu ch. 5 | how is a limit built in?        |
| PyTorch   | how is a cost differentiated?   |

# The Kinematic Bicycle Model

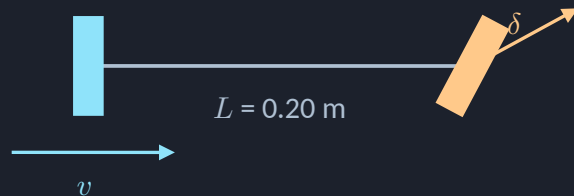
**Kinematic bicycle model** — the car reduced to two wheels: a speed and a steering angle in, position and heading out, with the wheelbase as its only parameter. Kinematic means no forces: it assumes the tyres never slide, so it holds exactly up to the grip limit and not an inch past it.

## THREE LINES, ONE PARAMETER

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \frac{v}{L} \tan \delta$$

$v$  the speed,  $\theta$  the heading,  $\delta$  the steering angle,  $L = 0.20$  m the wheelbase: the model's only parameter

### THE CAR, REDUCED



two wheels stand for four

the rear drives, the front turns

one parameter: the wheelbase

## BEFORE THE MODEL IS USED

**the wheelbase** 0.20 m, measured once

**command to angle** not linear: calibrated

**how** steady circles, radius taped

**skip it** and every result is in doubt

**suspension** real shocks: 'rigid' is false

*The Pro has oil-filled shocks, so weight transfer is visible: the model is knowingly simplified.*

# The Grip Limit

**Grip limit** — the hard ceiling the tyres set: one budget of force,  $\mu mg$ , that cornering and braking share, so the speed through a bend of curvature  $\kappa$  cannot pass the square root of  $\mu g / \kappa$ . A penalty is a preference an optimiser may trade away; on a car, trading it away is a crash.

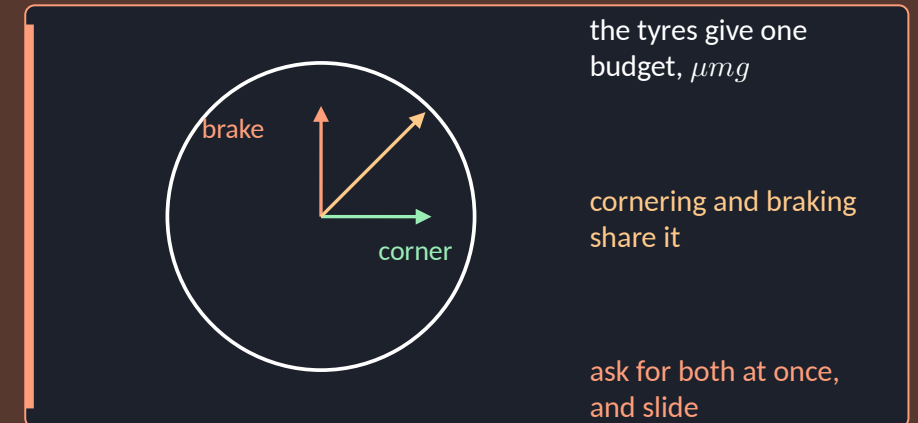
## THE CEILING, POINT BY POINT

$$v(s) \leq v_{lim}(s) = \min\left(\sqrt{\frac{\mu g}{\kappa(s)}}, v_{max}\right)$$

$\kappa$  the curvature of the course,  $\mu$  the tyre-floor friction,  $v_{max}$  the capped throttle's top speed

**TALK • 2 MIN** How would you explain, in your own words, the difference between a constraint in the loss and a constraint in the architecture? Then, in the drawing on the right, what happens to the cornering budget while the car brakes?

## ONE BUDGET OF FORCE



**soft: a penalty**

traded when the loss prefers

**hard: a layer**

cannot be traded at all

*L7.1's hard enforcement, with a crash as the stake.*

# Where the Joules Go

**Lap energy** — what the pack pays for one lap. The motor supplies rolling resistance, drag and acceleration; the current that takes is clamped at zero, because this drivetrain does not regenerate; and the power splits into winding heat, traction, and the computer's constant draw.

## THE FORCE, THE CURRENT, THE POWER

$$F = c_{rr} m g + \frac{1}{2} \rho C_d A v^2 + m v \frac{dv}{ds}, \quad i = \max\left(\frac{F}{K}, 0\right)$$

$$P = i^2 R_a + K i v + P_{hotel}$$

$c_{rr}$  rolling resistance,  $C_d A$  drag area,  $m v dv/ds$  acceleration,  $K$  newtons per ampere; braking draws no current

$i^2 R_a$  winding heat,  $K i v$  traction,  $P_{hotel} = 7 \text{ W}$ : the Nano, the camera, the display, paid for the whole lap

winding  $i^2 R_a$

traction  $K i v$

hotel 7 W

grows with the current squared

what moves the car

paid for the whole lap

## THE PLACEHOLDER CAR, FLAGGED

|                  | value                 |
|------------------|-----------------------|
| mass             | 1.05 kg               |
| friction $\mu$   | 0.65                  |
| rolling $c_{rr}$ | 0.02                  |
| motor $K, R_a$   | 1.0 N/A, 1.5 $\Omega$ |
| electronics      | 7 W, the whole lap    |
| course           | 18 m, three corners   |

Placeholders of the right order, not measurements - `problem.py` marks every one. No result is your car's until the on-car notebook has replaced them.  $c_{rr}$  includes the transmission: the 4WD's drag is absorbed into it.

# The Cheapest Lap

**Lap cost** — the energy of the lap plus the competition's price on its time, five joules a second. Driving slowly pays the electronics for longer; driving fast pays in winding heat and in the kinetic energy thrown away before each corner. So the cheapest lap is neither the slowest nor the fastest.

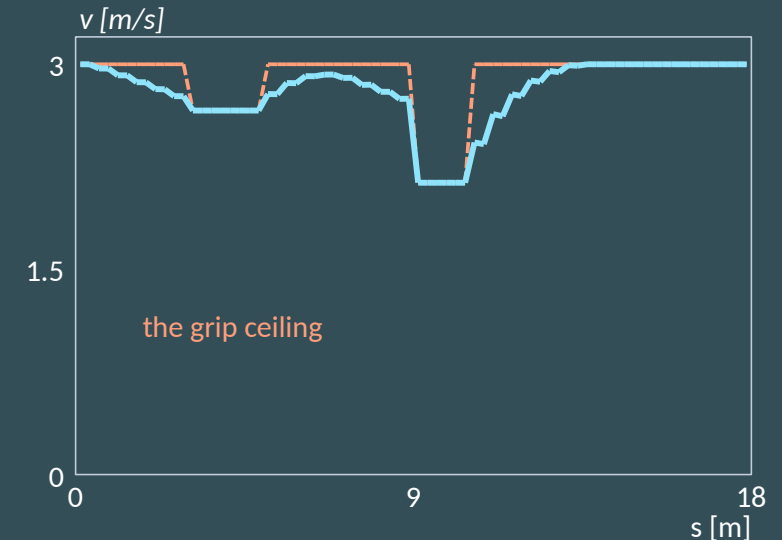
## THE COST, AND WHAT THE OPTIMISER FOUND

$$J = E + wT = \oint \frac{P(v)}{v} ds + w \oint \frac{ds}{v}, \quad w = 5 \text{ J/s}$$

$E$  the lap's energy,  $T$  its time,  $w$  what the competition pays per second saved

On the placeholder car the classical answer is 55.60 J in 6.49 s,  $J = 88.03$ : with 7 W of electronics and 5 J/s, time dominates, so the best lap runs close to the ceiling and brakes smoothly into each corner - the smooth profile is the cheap one.

## THE OPTIMISED SPEED, EX\_11.2'S OWN



*Dashed: the ceiling. Solid: the optimum, braking early and smoothly.*

# The Coast-Down

**Coast-down** — the identification run that costs nothing: cut the throttle and log the speed. No current flows, so only rolling and drag slow the car, and  $c_{rr}$  is read from the decay. A steady cruise cannot do this: there  $c_{rr}$  and  $R_a$  trade off, and the fit looks good anyway.

## THE FREEWHEELING EQUATION, FITTED

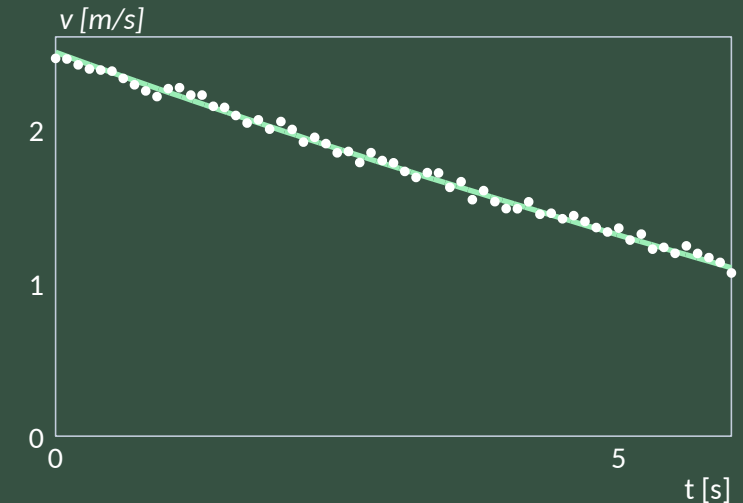
$$m \frac{dv}{dt} = -c_{rr} mg - \frac{1}{2} \rho C_d A v^2$$

$v(t)$  the speed of the freewheeling car: no current, so only rolling and drag slow it, and  $c_{rr}$  is read from the decay

### THE PHYSICS-INFORMED FIT

```
c_rr = torch.nn.Parameter(torch.tensor(0.01)) # the coefficient, trained with
the fit
res = m * grad(v, t) + c_rr * m * g + 0.5 * rho * CdA * v**2 # the residual
L = res.pow(2).mean() + W_D * (v - log_v).pow(2).mean() # physics + the
log
L.backward(); opt.step() # least squares agrees to 0.5 %
```

## THE SYNTHETIC LOG, AND ITS FIT



Dots: the log, made with  $c_{rr} = 0.0200$ ; least squares returns it, the physics-informed fit 0.0199. On the car the log is lap timing or the camera.

# The Limit as a Layer

**Bounded output layer** — hard enforcement for an optimiser: a sigmoid squeezes the output and the local ceiling scales it, so the speed can approach the grip limit and never pass it, whatever the weights do. The cost is then minimised by gradient descent through the lap's physics.

## THE SPEED PROFILE, BUILT SAFE

$$v(s) = v_{lim}(s) \sigma(\mathcal{N}(\cos ks, \sin ks)), \quad k = 1 \dots 6$$

$\sigma$  a sigmoid, between 0 and 1,  $\mathcal{N}$  the network on six harmonics of the closed lap:  $v$  can approach the limit, never pass it

### THE LAST LAYER

```
net = MLP(n_in=12, n_hidden=32, n_layers=3)    # six harmonics in: smooth,
v_close_lim * torch.sigmoid(net(h))           # can approach the limit, never
pass it
```

**TALK • 2 MIN** How would you decide, in your own words, when an optimisation needs a network at all? Then, from the table on the right, what did the six harmonics buy that a plain sigmoid network did not?

## AGAINST THE CLASSICAL OPTIMISER

|                  | L-BFGS-B | network         |
|------------------|----------|-----------------|
| energy           | 55.60 J  | 55.66 J         |
| lap time         | 6.49 s   | 6.48 s          |
| cost $J$         | 88.03 J  | 88.08 J         |
| apart            | —        | 0.05 %          |
| over the ceiling | never    | never, by build |

*288 bounded unknowns against a  $3 \times 32$  network on six harmonics of the closed lap. The earlier notebook promised the network a few per cent worse; with the harmonics and L-BFGS it is 0.05 %. Here the classical optimiser is excellent - the network's case is built-in safety and reuse, not victory.*

# The Efficiency Trophy

**Efficiency trophy** — the block's competition: six cars, three clean laps each, two boards. Board one is the cheapest lap by measured cost; board two the best energy prediction, committed before the run - a model is judged by what it said in advance, not afterwards.

## SIX CARS, TWO LEADERBOARDS

|                    |                                                        |
|--------------------|--------------------------------------------------------|
| <b>board one</b>   | the cheapest lap: least cost $J$ over three clean laps |
| <b>board two</b>   | the best prediction: committed before the run          |
| <b>clean</b>       | on the line, no cones, no hands                        |
| <b>measured by</b> | the car's own current sensor                           |

## WHAT EACH BOARD REWARDS

|                     |                               |
|---------------------|-------------------------------|
| <b>board one</b>    | driving and optimisation      |
| <b>board two</b>    | modelling and identification  |
| <b>one car</b>      | can win both                  |
| <b>the write-up</b> | says which numbers were yours |

*The study notebook's placeholder car scores nothing here: every parameter on the boards is one your team measured.*

# Advanced Topics

## RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

| topic                                   | what it adds                         | read            | where it makes a difference     |
|-----------------------------------------|--------------------------------------|-----------------|---------------------------------|
| <a href="#">the motor's own model</a>   | back-EMF, torque constant, winding   | Waveshare wiki  | where $K$ and $R_a$ come from   |
| <a href="#">the INA219's arithmetic</a> | shunt samples into a lap's joules    | Waveshare wiki  | the on-car measurement          |
| the pack under load                     | voltage sag, internal resistance     | Incropera ch. 1 | long runs; the battery of L10.1 |
| model predictive control                | the plan re-solved as the car drives | Liu ch. 5       | beyond a fixed profile          |
| weight transfer                         | grip that moves between the axles    | White ch. 2     | braking into a corner           |

### WHY THEY ARE LEFT OUT

Ex\_11.2 optimises one speed profile on a known course, scored against its own classical optimiser, so none of them is needed to do it. The first two rows are the on-car notebook's territory; the last is why the bicycle model is knowingly simplified.

# Exercise 11.2

## ONE STUDY NOTEBOOK, AND THE CAR

- 1 - 3 — the car's data, the force, the power, the grip limit
- 4 — the classical optimiser, one bounded unknown per step
- 5 — the network with the limit as its last layer; the two compared
- 6 — the rolling coefficient from a coast-down, two ways
- 7 - 9 — the report; then Ex11.2\_on\_the\_car: sensor, laps, boards



## THE QUESTION EACH PART ANSWERS

- 1 - 3 what does a lap pay for?
- 4 what is the classical answer?
- 5 can a layer keep the car safe?
- 6 what does a coast-down tell?

## REPORT QUESTION, TO CONCLUDE

On the car you commit an energy prediction before the scored laps, then measure. How far off was your prediction, which placeholder was most wrong, and what would you measure next to close the gap?

# Key Takeaways

## VEHICLE MODEL

|                        |                                    |
|------------------------|------------------------------------|
| <b>what it is</b>      | a bicycle model, one parameter     |
| <b>function</b>        | commands to a path, up to grip     |
| <b>train, tune</b>     | wheelbase measured, map calibrated |
| <b>vs the full car</b> | no forces, no sliding, knowingly   |
| <b>used for</b>        | the lap's kinematics and ceiling   |

## ENERGY

|                           |                                   |
|---------------------------|-----------------------------------|
| <b>what it is</b>         | winding heat, rolling, hotel load |
| <b>function</b>           | the cost $J = E + wT$ of a lap    |
| <b>train, tune</b>        | $c_{rr}$ from a coast-down        |
| <b>vs the slowest lap</b> | time is paid too: a best inside   |
| <b>used for</b>           | the committed prediction, scored  |

*Next — L12.1: the grid, where the network estimates what the meters cannot see.*

# Questions

## 1 – 2 • VEHICLE MODEL

### 1 What is the kinematic bicycle model, and what must be measured before it is used?

The car reduced to two wheels: speed and steering angle in, position and heading out, with the wheelbase  $L$  as its only parameter. The map from steering command to angle is not linear and is calibrated by measurement, or every later result is in doubt. *Ex\_11.2 on the car*

### 2 What is the grip limit, and why is it enforced hard?

The tyres give one budget of force, which braking and cornering share, so the speed in a bend of radius  $R$  cannot pass  $\sqrt{(\mu g R)}$ . A penalty is a preference the optimiser may trade away; a bounded layer cannot be traded, and on a car the difference is a crash. *Ex\_11.2 section 5; L7.1*

## 3 – 4 • ENERGY

### 3 Where does the energy go, and why is the cheapest lap not the slowest?

Into winding loss, which grows with the square of the current, into rolling resistance, and into the computer, which draws a constant power for as long as the lap takes. A slow lap pays the constant load longer, so the optimum lies in between, at a smooth speed profile. *Ex\_11.2 sections 4 – 5*

### 4 How are the model's parameters identified, and what makes the result checkable?

From an excitation run with steps, ramps and a coast-down; a steady cruise cannot separate the armature resistance from the rolling resistance, though the fit looks good. The current sensor measures the energy of a lap, so the prediction is compared with a measurement. *Ex\_11.2 section 6; L10.1*

NEXT

## L12.1 — Power Grid Stability Estimation

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The last block: the power grid, where a network proposes the state and the power flow equations judge it - with no true state to train on.

### BEFORE THE NEXT SESSION

|             |                                  |
|-------------|----------------------------------|
| read        | Liu ch. 5                        |
| run         | Ex_11.2's study notebook, 1 to 6 |
| bring       | your lap cost, and the 0.05 %    |
| for the lab | a working Ex_11.1 policy         |

*Block eleven is finished: a car that sees, and a car that knows what a lap costs.*